TAITIEN
Our Performance Your Reputation

# 100MHz *Low Noise/Low G-Sensitivity* OCXO
# NF-100M-6800 series

## NF-100M-6800 Series in 20.7x13.1mm DIP package

NF-100M-6800 series is a 100.000 MHz high performance (VC)OCXO offering low phase noise(LPN), low G sensitivity(LGS) and tight frequency stability down to ±50ppb(-20ºC to +70ºC). The part comes in a small hermetically sealed through hole package which makes it suitable for humid environmental conditions




RoHS Compliant Standard

### FEATURES

- ***Low Phase Noise & Low G-Sensitivity***
- Small Hermetically Sealed Package
- Tight Frequency Stability
- Low Power Consumption
- Fast Warm-up Time
- Electrical Frequency Tuning Input
- RoHS-Compliant (lead-free)

### APPLICATIONS

- Instrument Reference
- Satellite Communication
- Clock Reference for Microwave Signal Source
- Test & Measurement Equipment
- Avionics Systems
- Radar Systems

## ELECTRICAL SPECIFICATIONS

***Test conditions: VDC = +5 V; VCO = +2.25 V; at +25 ± 3°C unless otherwise identified***

### 1. OUTPUT (PIN = "R.F. OUTPUT")

| | Parameter | Min. | Typ. | Max. | Unit | Test Condition |
|---|---|---|---|---|---|---|
| 1.1. | Frequency (Fo) | | 100.000000 | | MHz | |
| 1.2. | Initial Accuracy | -0.2 | | +0.2 | ppm | @ +25 ±1°C |
| | | | | | | after turn on power 30 minutes |
| | | | | | | Vco=+2.25V |
| 1.3. | Waveform | | Sine wave/ HCMOS | | | |
| 1.4. | | | Sine wave | | | |
| 1.4.1. | Level | +8 | | | dBm | |
| 1.4.2. | Load | | 50 | | Ω | |
| 1.4.3. | Harmonics | | | -30 | dBc | |
| 1.4.4. | Spurious | | | -80 | dBc | |
| 1.5. | | | HCMOS | | | |
| 1.5.1. | Level High | 3.6 | | | V | |
| 1.5.2. | Level Low | | | 0.4 | V | |
| 1.5.3. | Load | | 1 kOhm // 10 pF | | | |
| 1.5.4. | Duty Cycle | 40% | | 60% | | |
| 1.5.5. | Rise/Fall time | | | 3 | ns | 20% to 80% |

## 2. FREQUENCY STABILITY

| | Parameter | Min. | Typ. | Max. | Unit | Test Condition | |
|---|---|---|---|---|---|---|---|
| 2.1. | Ambient | ±50, ±100, ±200 | | | ppb | referred to 25°C | **Refer to Table 1 : Ordering Information** |
| | | -20ºC ~ +70ºC<br>-40ºC ~ +85ºC | | | °C | | |
| 2.2. | Aging | | | | | | |
| 2.2.1. | Daily | -5 | | +5 | ppb | after 30 days | |
| 2.2.2. | Yearly | -500 | | +500 | ppb | | |
| 2.2.3. | 10 Years | -1.7 | | +1.7 | ppm | | |
| 2.3. | Voltage | -30 | | +30 | ppb | ±5% change | |
| 2.4. | Short term | | | 0.05 | ppb | root Allan variance for τ=1 sec | |
| 2.5. | Load | -10 | | +10 | ppb | ±5% change | |
| 2.6. | Warm-up | -100 | | +100 | ppb | in 5 minutes @ +25 ±1°C | referred to 1 hour |
| 2.7. | G-Sensitivity (each axis) | | | 1 | ppb/g | | |

| | | Option A | Option B | | Refer to Table 1 : Ordering Information |
|---|---|---|---|---|---|
| 2.8. | Phase Noise (Max.) | -97 | -102 | dBc/Hz | @ 10Hz |
| | | -130 | -135 | dBc/Hz | @ 100Hz |
| | | -160 | -162 | dBc/Hz | @ 1KHz |
| | | -173 | -175 | dBc/Hz | @ 10KHz |
| | | -177 | -177 | dBc/Hz | @ 100KHz |
| | | -178 | -178 | dBc/Hz | @ 1MHz |

## 3. ELECTRICAL FREQUENCY ADJUSTMENT (PIN = "VCO INPUT")

| | Parameter | Min. | Typ. | Max. | Unit | Test Condition |
|---|---|---|---|---|---|---|
| 3.1. | Tuning Range | ±2 | | | ppm | Referenced to frequency at nominal Center Voltage |
| 3.2. | Control Voltage | 0 | | +5.0 | V | |
| 3.3. | Slope | Positive | | | | |
| 3.4. | Center Voltage | | +2.5 | | V | |
| 3.5. | Linearity | -10 | | +10 | % | |

## 4. INPUT POWER (PIN = "+VDC")

| | Parameter | Min. | Typ. | Max. | Unit | Test Condition |
|---|---|---|---|---|---|---|
| 4.1. | Voltage | +4.75 | +5 | +5.25 | V | |
| 4.2. | Current | | | | | |
| 4.2.1. | Steady State | | | 240 | mA | |
| 4.2.2. | During Warm-Up | | | 600 | mA | |

## 5. ENVIRONMENTAL

| | Parameter | Reference Std. | Test Condition |
|---|---|---|---|
| 5.1. | Operable Temperature | -40°C to +85°C | Note 1 |
| 5.2. | Storage Temperature | -45°C to +90°C | |
| 5.3. | Humidity | MIL-STD-202, Method 103<br>Test Condition A | 95% RH @ +40°C,non-condensing,240 hours |
| 5.4. | Vibration (non-operating) | MIL-STD-202, Method 201 | 0.06" Total p-p, 10 to 55 Hz |
| 5.5. | Shock (non-operating) | MIL-STD-202, Method 213,<br>Test Condition J | 30g, 11ms, half-sine |

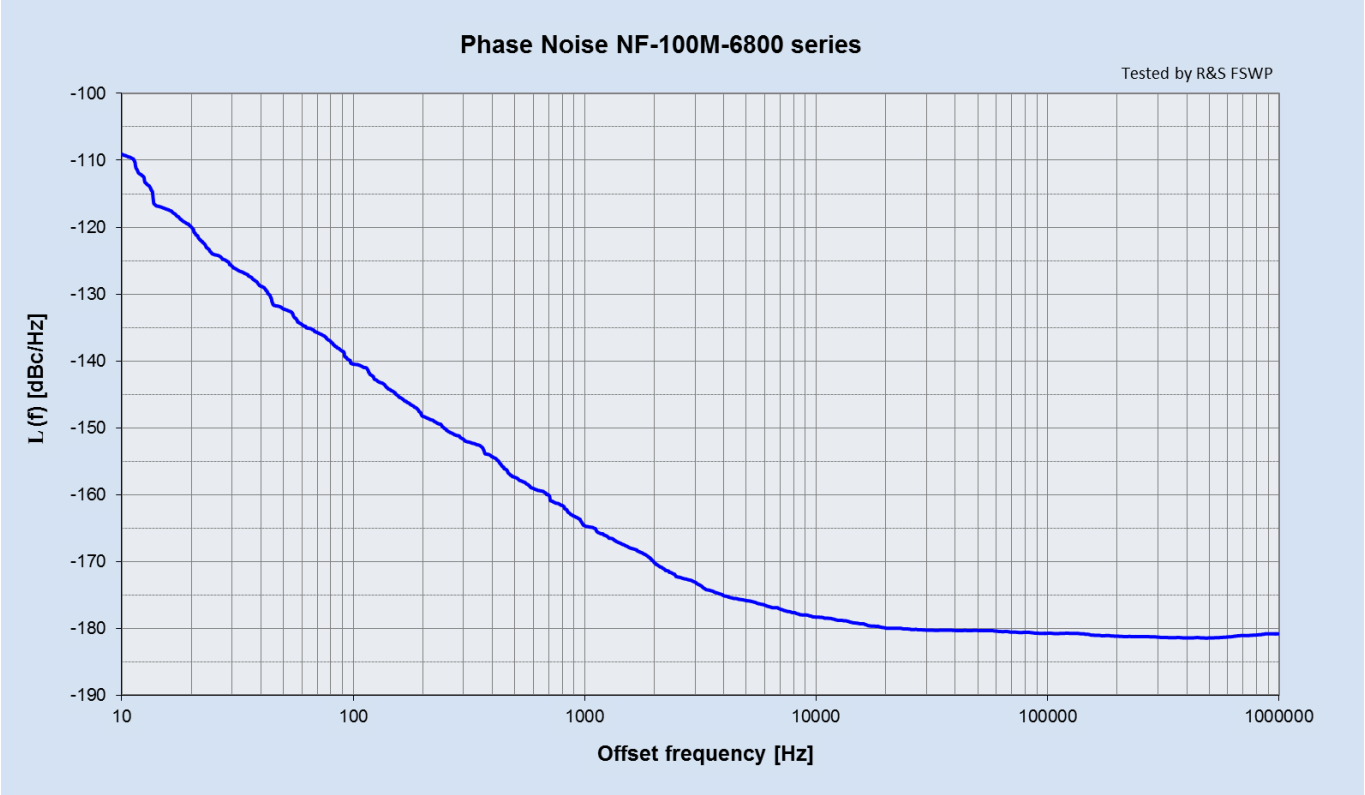
**Note 1 :** Output maintained over this temperature range. Other requirements of this specification may not be met when operating outside the temperature range in 2.1.

## Table 1 : ORDERING INFORMATION

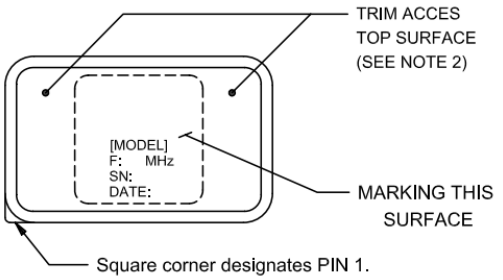
| Ambient Temp. (°C) / Option | | Output | Phase Noise Option | |
|---|---|---|---|---|
| | | | A | B |
| -20°C ~+70°C | ±100 ppb | SineWave | NF-100M-6800 | NF-100M-6801 |
| | ±50 ppb | | NF-100M-6810 | NF-100M-6811 |
| -40°C ~+85°C | ±200 ppb | | NF-100M-6820 | NF-100M-6821 |
| | ±100 ppb | | NF-100M-6830 | NF-100M-6831 |
| -20°C ~+70°C | ±100 ppb | HCMOS | NF-100M-6850 | NF-100M-6851 |
| | ±50 ppb | | NF-100M-6860 | NF-100M-6861 |
| -40°C ~+85°C | ±200 ppb | | NF-100M-6870 | NF-100M-6871 |
| | ±100 ppb | | NF-100M-6880 | NF-100M-6881 |

Other specifications may be available upon request.

## Phase Noise Test Data

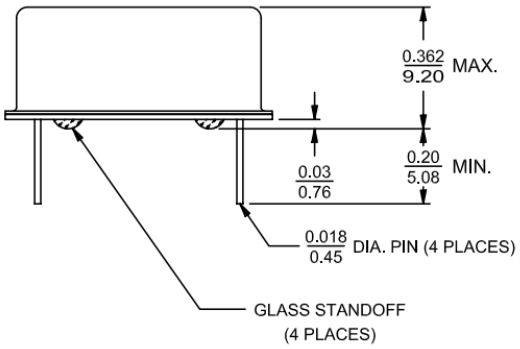


# OUTLINE DRAWING




(VIEW FROM TOP)

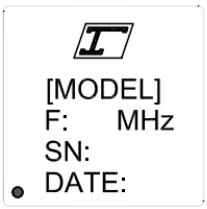
| PIN CONNECTIONS | |
|---|---|
| PIN | FUNCTION |
| 1 (See Note 1) | VCO INPUT or NOT CONNECTED |
| 7 | 0 VOLTS & CASE |
| 8 | R.F. OUTPUT |
| 14 | +VDC |

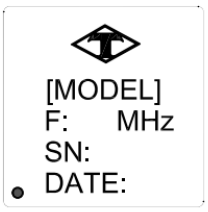


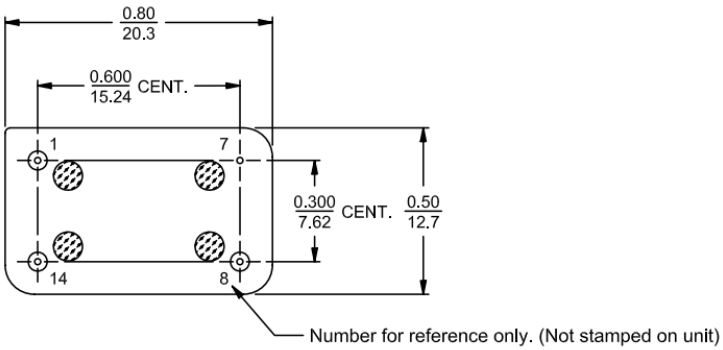

Notes:

1. If the specification does not specify parameters for PIN 1 then PIN 1 is not internally CONNECTED.
2. If the specification does not specify parameters for "MECHANICAL FREQUENCY ADJUSTMENT" then there is no Trim Access.

MARKING




(VIEW FROM BOTTOM)

INCH / mm (REFERENCE ONLY)

| REV. | DATE | DESCRIPTION | ECO.NO. | DRAWN | DESIGNED | CHECKED | APPROVED | SCALE |
|---|---|---|---|---|---|---|---|---|
| 1.0 | 07/03/2017 | | N003000 | CYHu | CL | CL | IMYeh | 2:1 |
| | | | | | | | | DATE |
| | | | | | | | | 07/03/2017 |

TAITIEN ELECTRONICS(NANJING)CO.,LTD.

TOLERANCES:
UNLESS OTHERWISE SPECIFIED:
ANGLES: ±1 DEGREE
FRACTIONS:±1/32 INCH
DECIMALS: .XX±.015, .XXX±.010 INCH

M-NF-002-N

REV. 1.0